

Netaji Subhas University of Technology



CSE – 3

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MiniOCR: VLM based OCR architecture

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INTRODUCTION

Optical Character Recognition [\[1\]](#) (OCR) is a technology used to identify and convert textual information present in images into machine-readable representations. It forms an important component of systems that process scanned documents, photographs, forms, handwritten notes, industrial labels, and other visual sources.

By converting visual text into digital text, OCR enables downstream operations such as searching, indexing, editing, classification, and automated information extraction.

Although the fundamental objective of OCR is text recognition, practical OCR systems must deal with a wide range of visual and linguistic variations. Differences in font, character size, writing style, image quality, illumination, background, orientation and document layout can significantly affect recognition. Handwritten and degraded text can be particularly challenging because individual characters may be ambiguous even when the surrounding context provides enough information to determine the intended word.

Recent developments in Transformer-based vision and language models have provided an alternative approach to OCR. Vision-Language [\[2\]](#) (VLMs) are designed to process visual information together with language representations, allowing visual features to be interpreted in a broader linguistic context.

When characters are partially degraded or unclear, VLMs can use contextual information from neighboring characters and words to determine what the word is most likely to be. Similarly, a model that processes a complete page can potentially use spatial relationships between different regions to handle multi-column documents, tables, and irregular layouts more effectively.

Despite these advantages, VLM-based OCR introduces another challenge: computational complexity. Large multimodal models may contain a substantial number of parameters and can require considerable computational resources for both training and inference[8] Such requirements can make them unsuitable for environments where memory, processing power or energy consumption is constrained.

This MiniOCR project investigates whether a compact VLM-based OCR architecture can provide useful OCR performance while remaining suitable for resource-constrained or edge-device deployment. The project targets a model of approximately 20 million parameters and explores a compact Vision Transformer (ViT), an OCR-specific Q-Former, a character-aware tokenizer, and an autoregressive GPT-style decoder.

The planned pipeline includes image preprocessing, patch embedding, visual encoding, spatial encoding, visual projection, OCR-specific visual compression, reading-order-aware attention, tokenization, and text generation.

MOTIVATION

The central motivation is the gap between the flexibility of VLM-based OCR and their computational requirements. There are several advantages of VLM-based OCR, including contextual inference for degraded characters, improved handling of multi-column pages, tables and irregular layouts, and the possibility of supporting multiple languages and scripts with a single transformer-based system.

At the same time, we have identified computational cost and training data requirements as major limitations. Large VLM-based OCR systems may require substantial hardware, while small models trained on limited data can underperform.

MiniOCR therefore focuses on the question of whether a small model can provide adequate OCR capability within edge-device resource constraints.

A compact architecture also provides an opportunity to study which components are essential for multimodal OCR. Instead of relying on a very large general-purpose VLM, the proposed design allocates model capacity specifically to visual encoding, OCR-specific feature compression, reading-order handling, and text generation.

LITERATURE SURVEY

We have surveyed several approaches relevant to OCR and VLM-based text recognition, given as follows:

TrOCR: Transformer-Based Optical Character Recognition with Pre-Trained Models[\[7\]](#)

TrOCR uses an image Transformer encoder and a text Transformer decoder. Image patches are represented by the encoder, and the decoder generates word-piece sequences using the visual representations and previous predictions. The surveyed document notes that this approach performs strongly across printed, handwritten, and scene-text recognition tasks.

DTrOCR: Decoder-Only Transformer for Optical Character Recognition[\[5\]](#)

DTrOCR uses a decoder-only Transformer for text recognition. The input image is represented as a patch sequence, and recognition is performed autoregressively. The surveyed work also uses pretraining and fine-tuning to reduce computational requirements. The MiniOCR design draws on the lightweight decoder-only direction while adapting the architecture to include a dedicated OCR visual compression stage.

Deep Learning-Based OCR for Robust Real-World Conditions[\[4\]](#)

The surveyed work examines Bidirectional Long Short-Term Memory (Bi-LSTM) networks for OCR under varying fonts, sizes, styles, and handwriting. The document identifies Bi-LSTM as a viable architecture for irregular, noisy, or handwritten text and treats this as one possible source of design lessons for the proposed system.

Attention-Based Deep Learning Algorithm in NLP for OCR[3]

The surveyed CNN-RNN architecture adds attention and uses multiple levels of segmentation, including line, word, and character segmentation. The reported work uses a 65K handwritten-character dataset and reports high accuracy. Its segmentation strategy provides an alternative to end-to-end recognition and highlights the importance of spatial structure in OCR.

OCR-Augmented GPT for Accurate Text Extraction in Industrial Environments[6]

This work combines outputs from existing OCR tools with a GPT layer and the original images. The surveyed approach demonstrates the value of combining preliminary OCR information with contextual language understanding for degraded industrial text. For MiniOCR, the relevant design lesson is the potential benefit of multimodal/contextual processing for reducing OCR inaccuracies.

PROBLEM STATEMENT

Existing OCR systems can face difficulties when input text is degraded, handwritten, noisy, multilingual, or arranged in complex layouts. VLM-based OCR can address some of these challenges through joint visual and language reasoning, but such systems can be computationally expensive and may require large datasets.

The problem addressed by MiniOCR is therefore to design and investigate a compact VLM-based OCR architecture that can process an input image end-to-end and generate recognized text while maintaining a substantially smaller computational footprint than large multimodal models.

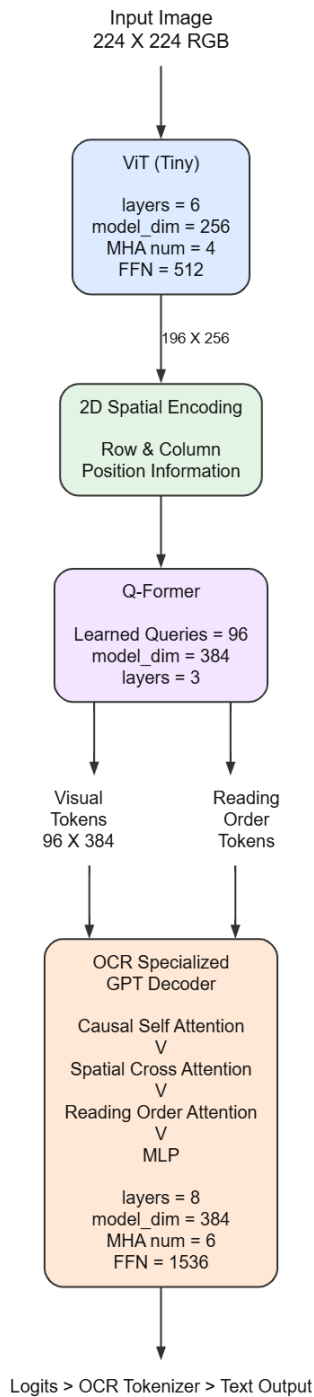
The proposed architecture targets approximately 20 million parameters and is intended to explore whether a compact combination of a Vision Transformer, OCR-specific visual compression, reading-order-aware attention, and an autoregressive text decoder can provide useful OCR performance under constrained resources.

OBJECTIVE AND METHODOLOGY

Objectives

- Design a compact VLM-based OCR architecture.
- Introduce an OCR-specific Q-Former to compress spatial visual features into a smaller visual memory.
- Incorporate spatial and reading-order information so that complex document layouts can be represented.
- Evaluate and benchmark the proposed model using Character Error Rate (CER) and Word Error Rate (WER).
- Investigate suitability for resource-constrained or edge-device use.

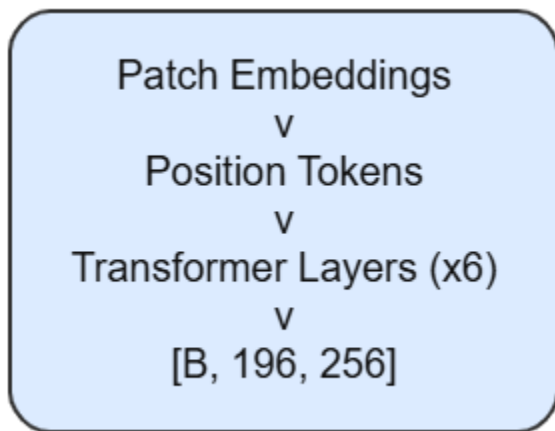
Proposed architecture



Input preprocessing converts a 224×224 RGB image into normalized tensors with shape $[B, 3, 224, 224]$. The image is divided into 16×16 patches,

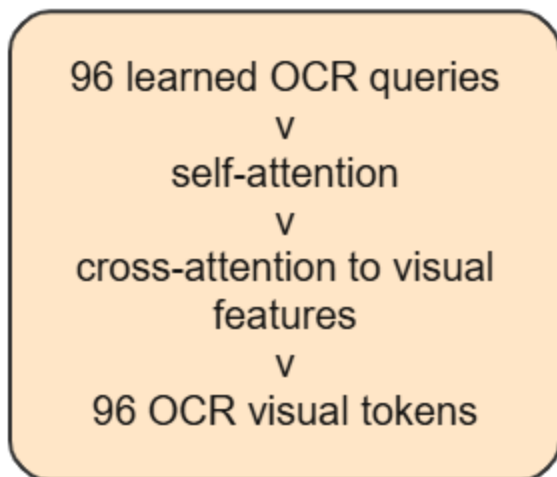
resulting in $14 \times 14 = 196$ patches. Each patch is embedded into a 256-dimensional representation.

The ViT encoder contains six layers with four attention heads, a hidden dimension of 256, an FFN dimension of 512, and dropout of 0.1. Each block uses self-attention, residual connections, LayerNorm, and an FFN.

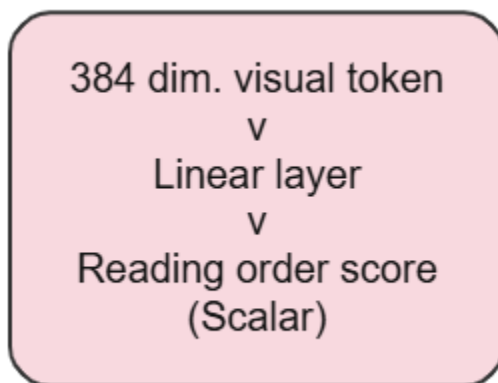


Spatial encoding adds row and column information to preserve the position of visual patches. The visual features are then projected from $[B, 196, 256]$ into $[B, 196, 384]$.

The OCR Q-Former compresses the 196 visual features into OCR-specific visual memory tokens. The proposed Q-Former uses 96 learned queries, hidden dimension 384, three layers, six attention heads, and an FFN dimension of 1024.

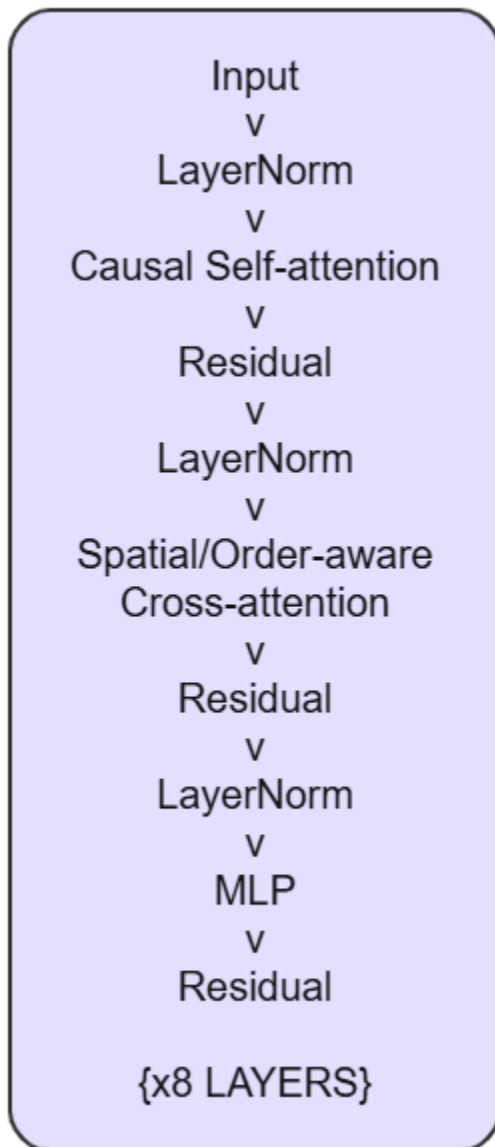


A reading-order mechanism is introduced to provide the visual representation with an explicit notion of document reading order. The proposed attention modifies the standard attention score with an order compatibility term. The document notes that this should remain a soft bias because documents may contain multiple columns, tables, and irregular layouts.



The OCR-specific tokenizer is planned as a character-aware BPE tokenizer with an initial vocabulary of approximately 5,000 units. Its training corpus includes letters, digits, punctuation, whitespace, OCR-specific symbols, frequent character sequences, and irregular characters.

The GPT-OCR decoder generates the recognized text autoregressively. The proposed decoder has eight layers, hidden dimension 384, six attention heads, FFN dimension 1536, and GELU activation. It uses causal self-attention together with visual cross-attention, with the latter incorporating the reading-order bias.



Dataset Preparation and Model Training

The current dataset consists of 30,000 (24,000 training/6,000 testing) synthetically generated OCR images, each of 300x300 dimensions. The dataset was generated using dedicated PyPi libraries[\[9\]](#) This dataset is being used for pretraining the VLM-OCR pipeline and evaluating the first-stage training run.

The model will then be pretrained on the curated dataset and fine-tuned for domain-specific tasks such as handwriting or structured document OCR.

Evaluation

Evaluation is planned on held-out data from each target domain. The primary metrics identified in the project document are Character Error Rate (CER) and Word Error Rate (WER).

Character Error Rate - The **Character Error Rate (CER)** is a metric used to evaluate the accuracy of text recognition systems at the character level. The formula for calculating CER is:

$$\text{CER} = (\text{S} + \text{D} + \text{I}) / \text{N} \times 100$$

Where:

S = Substitutions (characters recognized incorrectly)

D = Deletions (characters missing from output)

I = Insertions (extra characters added to output)

N = Total number of characters in the reference text

A lower CER value indicates better performance, with ~0% representing perfect accuracy.

Word Error Rate - The **Word Error Rate (WER)** is a metric used to evaluate the accuracy of text recognition systems at the word level. The formula for calculating WER is:

$$\text{WER} = (\text{S} + \text{D} + \text{I}) / \text{N} \times 100$$

Where:

S = Substitutions (words recognized incorrectly)

D = Deletions (words missing from output)

I = Insertions (extra words added to output)

N = Total number of characters in the reference text

A lower WER value indicates better performance, with ~0% representing perfect accuracy.

The results obtained from the model training and benchmarking will be used to further optimize and enhance the architecture, both during training and post-training.

SIMULATION PLATFORM AND REQUIREMENTS

Software Requirements

- Operating System: Windows
- Programming Language: Python 3.11
- Deep Learning Framework: PyTorch + TensorFlow
- CUDA / GPU software: 12.1+
- Supporting libraries: Transformers / torchvision / tokenizers / NumPy / etc.
- Development environment: VS Code

Hardware Requirements

- CPU: i7 11th gen
- RAM: 16 GB
- GPU: RTX 3050 (4GB VRAM)
- Storage: 100 GB

CONCLUSION

MiniOCR proposes a compact VLM-based OCR architecture designed around the practical tension between multimodal OCR capability and computational efficiency. The architecture combines a compact Vision Transformer, spatial encoding, an OCR-specific Q-Former, learned query tokens, reading-order-aware cross-attention, a custom character-aware BPE tokenizer, and a GPT-adjacent autoregressive decoder.

This project is motivated by the possibility that a substantially smaller model can retain useful contextual OCR capabilities while reducing the resources required for inference. The proposed evaluation using CER and WER will provide an empirical basis for determining how effectively the architecture performs on the selected OCR domains.

REFERENCES

1. Wang, Haifeng & Pan, Changzai & Guo, Xiao & Ji, Chunlin & Deng, Ke. (2021). From object detection to text detection and recognition: A brief evolution history of optical character recognition. WIREs Computational Statistics. 13. 10.1002/wics.1547.
2. <https://doi.org/10.48550/arXiv.2405.17247>
3. P. K. M, R. Logesh Babu, G. Sunil, J. Sampathkumar and S. Chakraborty, "Attention-Based Deep Learning Algorithm in Natural Language Processing for Optical Character Recognition," 2023 International Conference on Evolutionary Algorithms and Soft Computing Techniques (EASCT), Bengaluru, India, 2023, pp. 1-5, doi:10.1109/EASCT59475.2023.10393635.
4. A. Mishra, J. Sikdar and S. U. Kumar, "Deep Learning-Based Optical Character Recognition for Robust Real-World Conditions: A Comparative Analysis," 2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT), Kamand, India, 2024, pp. 1-7, doi: 10.1109/ICCCNT61001.2024.10726007.
5. Fujitake, Masato. (2023). DTrOCR: Decoder-only Transformer for Optical Character Recognition. 10.1109/WACV57701.2024.00784.
6. J. Park, J. Cho, S. Han and K. Kim, "OCR-Augmented GPT for Accurate Text Extraction in Industrial Environments," in IEEE Access, vol. 13, pp. 136226-136236, 2025, doi: 10.1109/ACCESS.2025.3594682.
7. Li, Minghao & Lv, Tengchao & Cui, Lei & Lu, Yijuan & Florencio, Dinei & Zhang, Cha & Li, Zhoujun & Wei, Furu. (2021). TrOCR: Transformer-based Optical Character Recognition with Pre-trained Models. 10.48550/arXiv.2109.10282.
8. J. Kim, C. -Y. Lee, C. -Y. Han, N. -H. Yoo and J. Yang, "Beyond Simple Character Recognition: A Comparative Study of Vision-Language Models and Dedicated OCR Systems in Edge Cases," in IEEE Access, vol. 14, pp. 94148-94166, 2026, doi: 10.1109/ACCESS.2026.3705442.
9. [ocr-data-toolkit · PyPI](#)